

Carpark

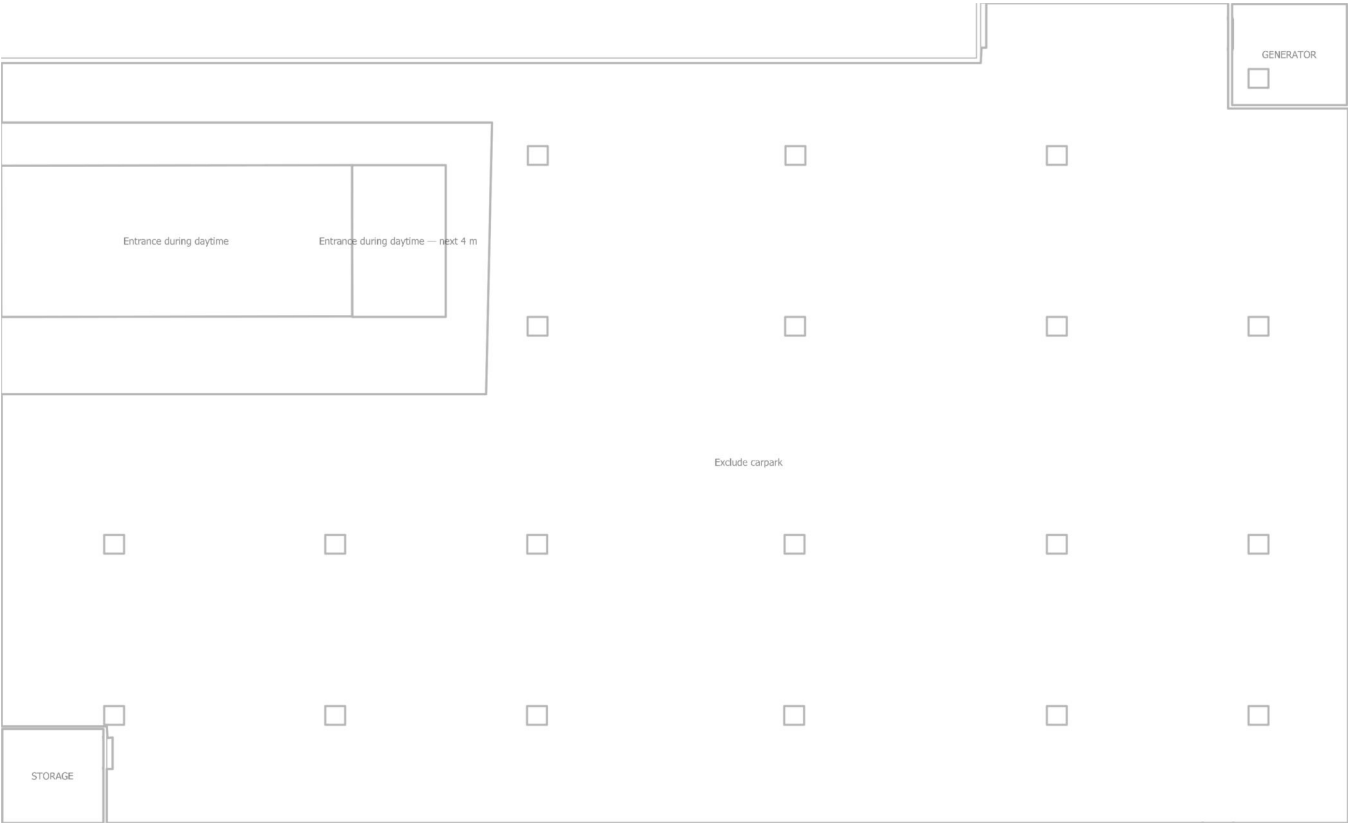
Luminaire list

Φ_{total} 382111 lm	P_{total} 2774.3 W	Luminous efficacy 137.7 lm/W	$\Phi_{\text{Emergency lighting}}$ 14587 lm	$P_{\text{Emergency lighting}}$ 47.7 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
53	ETAP	-	K212/3-A-D800A40 (LTRRevAC)	0.9 W	275 lm	312.5 lm/W
				 0.9 W	275 lm (100 %)	-
12	ETAP	K9R142/1-AX1		2.5 W	80 lm	32.0 lm/W
				 0.0 W	1 lm (1 %)	-
18	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	64.0 W	10782 lm	168.5 lm/W
88	Lena Lighting	100765	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)	14.9 W	1650 lm	110.7 lm/W
6	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR (Emergency light scene)

Room list



Carpark · GROUND FLOOR (Emergency light scene)

Room list

Entrance during daytime

P_{total}
4.5 W**A_{Room}**
96.98 m²**Lighting power density**
0.05 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
5	ETAP	-	K212/3-A-D800A40 (_LTRevAC)	0.9 W	275 lm (100 %)
1	ETAP	K9R142/1-AX1		0.0 W	1 lm (1 %)

Entrance during daytime — next 4 m

P_{total}
0.9 W**A_{Room}**
25.95 m²**Lighting power density**
0.03 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K212/3-A-D800A40 (_LTRevAC)	0.9 W	275 lm (100 %)

Exclude carpark

P_{total}
39.6 W**A_{Room}**
1625.11 m²**Lighting power density**
0.02 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
44	ETAP	-	K212/3-A-D800A40 (_LTRevAC)	0.9 W	275 lm (100 %)
9	ETAP	K9R142/1-AX1		0.0 W	1 lm (1 %)

Carpark · GROUND FLOOR (Emergency light scene)

Room list

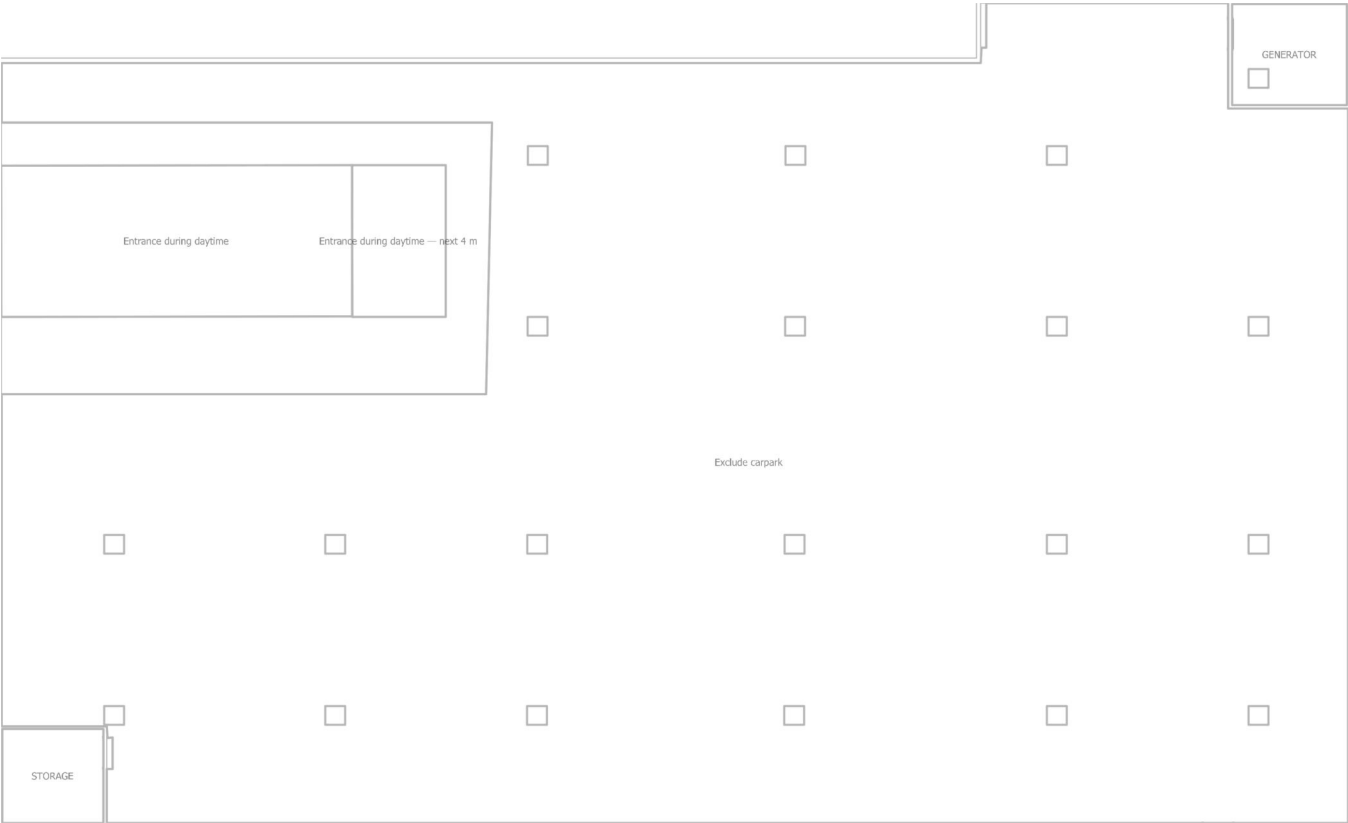
GENERATOR

 P_{total}
0.9 W A_{Room}
21.21 m²Lighting power density
0.04 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K212/3-A-D800A40 (_LTRevAC)	0.9 W	275 lm (100 %)
1	ETAP	K9R142/1-AX1		0.0 W	1 lm (1 %)

Carpark · GROUND FLOOR (Light scene 1)

Room list



Carpark · GROUND FLOOR (Light scene 1)

Room list

Entrance during daytime

P_{total} 1024.0 W	A_{Room} 96.98 m ²	Lighting power density 10.56 W/m ² = 1.28 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 825 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
16	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	64.0 W	10782 lm

Entrance during daytime — next 4 m

P_{total} 128.0 W	A_{Room} 25.95 m ²	Lighting power density 4.93 W/m ² = 1.07 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 459 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
2	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	64.0 W	10782 lm

Exclude carpark

P_{total} 1251.6 W	A_{Room} 1625.11 m ²	Lighting power density 0.77 W/m ² = 1.79 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 43.0 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
84	Lena Lighting	100765	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)	14.9 W	1650 lm

Carpark · GROUND FLOOR (Light scene 1)

Room list

GENERATOR

P_{total}
155.6 W**A_{Room}**
21.21 m²**Lighting power density**
7.34 W/m² = 3.61 W/m²/100 lx (Space)**E_{perpendicular} (Working plane)**
203 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
4	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	38.9 W	4550 lm

STORAGE

P_{total}
77.8 W**A_{Room}**
17.40 m²**Lighting power density**
4.47 W/m² = 2.93 W/m²/100 lx (Space)**E_{perpendicular} (Working plane)**
153 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
2	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	38.9 W	4550 lm

Carpark · GROUND FLOOR

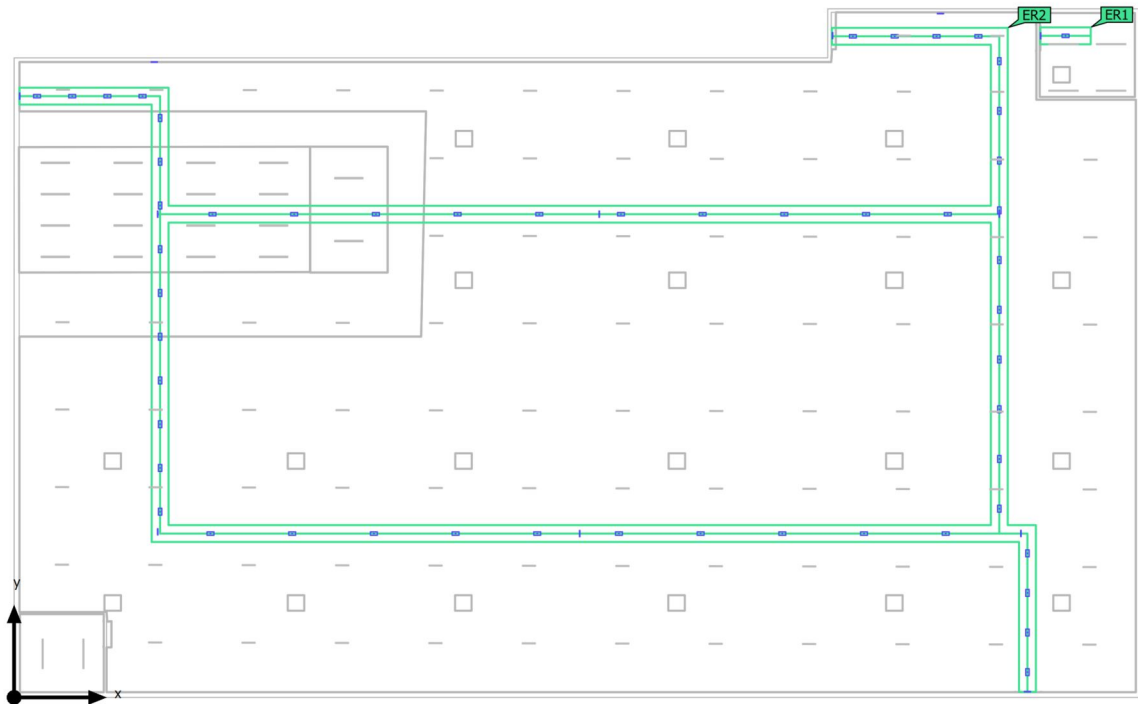
Luminaire list

Φ_{total} 382111 lm	P_{total} 2774.3 W	Luminous efficacy 137.7 lm/W	$\Phi_{\text{Emergency lighting}}$ 14587 lm	$P_{\text{Emergency lighting}}$ 47.7 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
53	ETAP	-	K212/3-A-D800A40 (LTRRevAC)	0.9 W	275 lm	312.5 lm/W
				 0.9 W	275 lm (100 %)	-
12	ETAP	K9R142/1-AX1		2.5 W	80 lm	32.0 lm/W
				 0.0 W	1 lm (1 %)	-
18	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	64.0 W	10782 lm	168.5 lm/W
88	Lena Lighting	100765	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)	14.9 W	1650 lm	110.7 lm/W
6	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR (Emergency light scene)

Calculation objects



Carpark · GROUND FLOOR (Emergency light scene)

Calculation objects

Escape routes

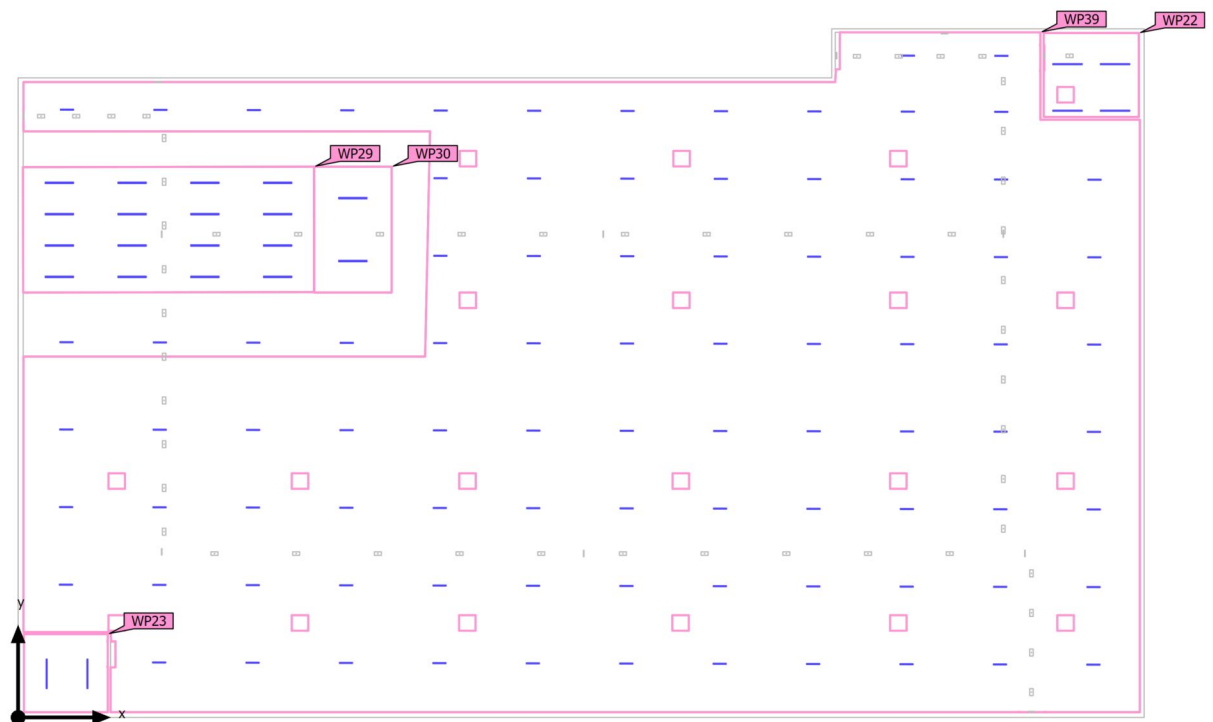
Properties	E_{min} Middle area (Target)	E_{max} Middle area	E_{min} Centerline (Target)	E_{max} Centerline	U_d (Target)	Index
Emergency route 4 Perpendicular illuminance (adaptive) Height: 0.000 m	3.99 lx (≥ 0.50 lx) ✓	7.37 lx	4.04 lx (≥ 1.00 lx) ✓	6.67 lx	0.61 (≥ 0.025) ✓	ER1
Emergency route 7 Perpendicular illuminance (adaptive) Height: 0.000 m	1.63 lx (≥ 0.50 lx) ✓	12.1 lx	1.64 lx (≥ 1.00 lx) ✓	11.5 lx	0.14 (≥ 0.025) ✓	ER2

Notes on planning:

The emergency lighting scene was calculated without reflection and taking into account the placed furniture.

Carpark · GROUND FLOOR (Light scene 1)

Calculation objects



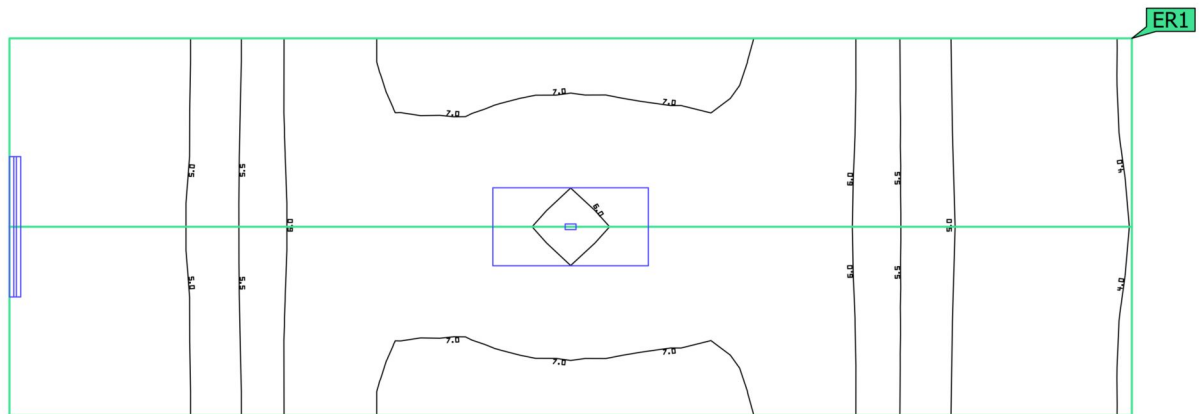
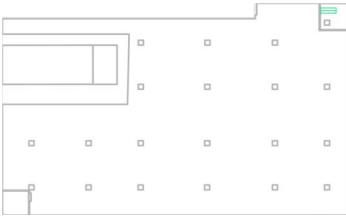
Carpark · GROUND FLOOR (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Entrance during daytime — next 4 m) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	459 lx (≥ 400 lx) ✓	201 lx	686 lx	0.44 (≥ 0.40) ✓	0.29	WP30
Working plane (Entrance during daytime) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	825 lx (≥ 800 lx) ✓	345 lx	1090 lx	0.42 (≥ 0.40) ✓	0.32	WP29
Working plane (Exclude carpark) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	43.0 lx (≥ 40.0 lx) ✓	15.9 lx	254 lx	0.37 (≥ 0.30) ✓	0.063	WP39
Working plane (GENERATOR) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	203 lx (≥ 160 lx) ✓	91.6 lx	266 lx	0.45 (≥ 0.40) ✓	0.34	WP22
Working plane (STORAGE) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.000 m	153 lx (≥ 80.0 lx) ✓	66.2 lx	225 lx	0.43 (≥ 0.40) ✓	0.29	WP23

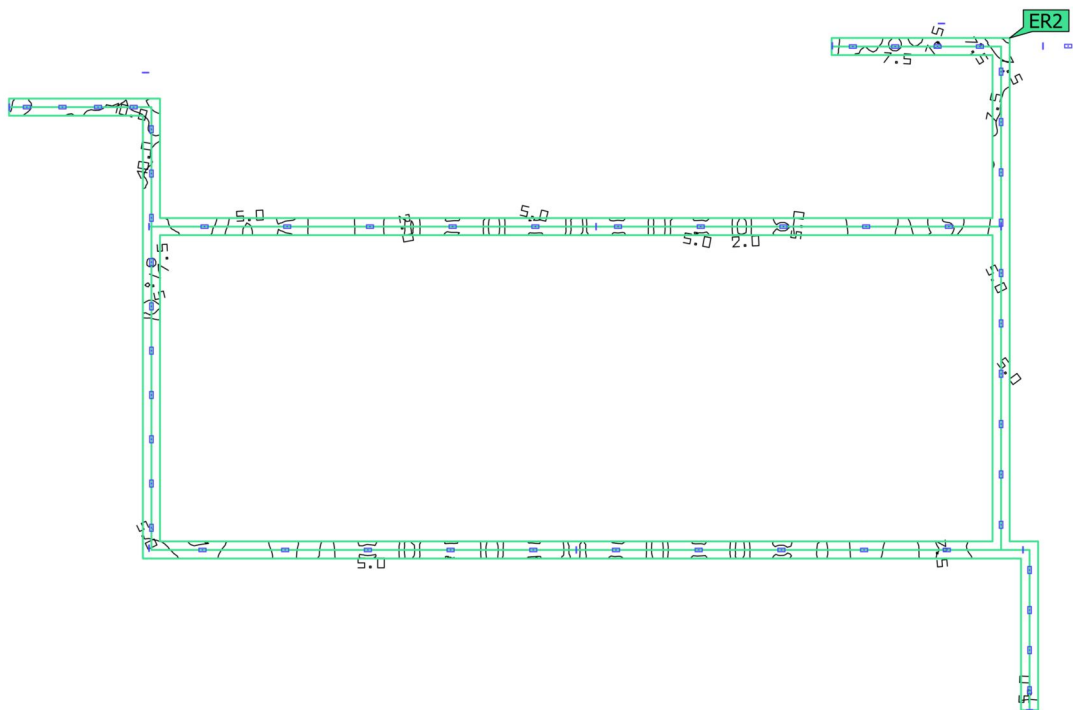
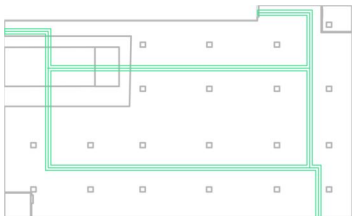
Carpark · GROUND FLOOR (Emergency light scene)

Emergency route 4

Properties	E_{min} Middle area (Target)	E_{max} Middle area	E_{min} Centerline (Target)	E_{max} Centerline	U_d (Target)	Index
Emergency route 4 Perpendicular illuminance (adaptive) Height: 0.000 m	3.99 lx (≥ 0.50 lx) ✓	7.37 lx	4.04 lx (≥ 1.00 lx) ✓	6.67 lx	0.61 (≥ 0.025) ✓	ER1

Notes on planning:
The emergency lighting scene was calculated without reflection and taking into account the placed furniture.

Carpark · GROUND FLOOR (Emergency light scene)
Emergency route 7

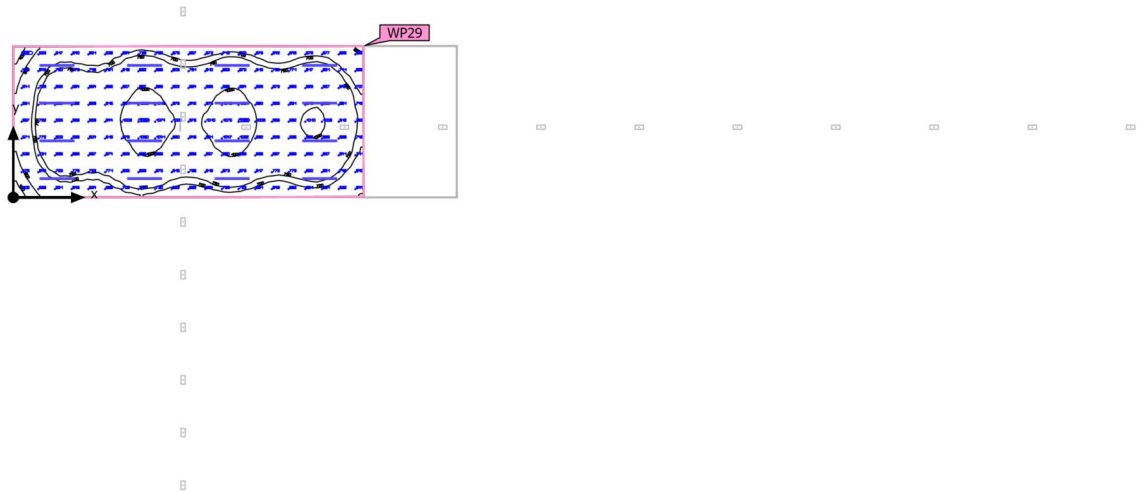


Properties	E_{min} Middle area (Target)	E_{max} Middle area	E_{min} Centerline (Target)	E_{max} Centerline	U_d (Target)	Index
Emergency route 7 Perpendicular illuminance (adaptive) Height: 0.000 m	1.63 lx (≥ 0.50 lx) ✓	12.1 lx	1.64 lx (≥ 1.00 lx) ✓	11.5 lx	0.14 (≥ 0.025) ✓	ER2

Notes on planning:
The emergency lighting scene was calculated without reflection and taking into account the placed furniture.

Carpark · GROUND FLOOR · Entrance during daytime (Light scene 1)

Summary



Ground area	96.98 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.62 (fixed)

Mounting height	2.800 m
Height _{Working plane}	0.000 m
Wall zone _{Working plane}	0.000 m

Carpark · GROUND FLOOR · Entrance during daytime (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	825 lx	$\geq 800 \text{ lx}$	✓	WP29
	$U_o (g_1)$	0.42	≥ 0.40	✓	WP29
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	30	≤ 25	✗	
Energy estimation ⁽²⁾	Consumption	1126 kWh/a	max. 3400 kWh/a	✓	
Space	Lighting power density	10.56 W/m ²	–		
		1.28 W/m ² /100 lx	–		

(1) Based on a rectangular space of 15.000 m x 6.466 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

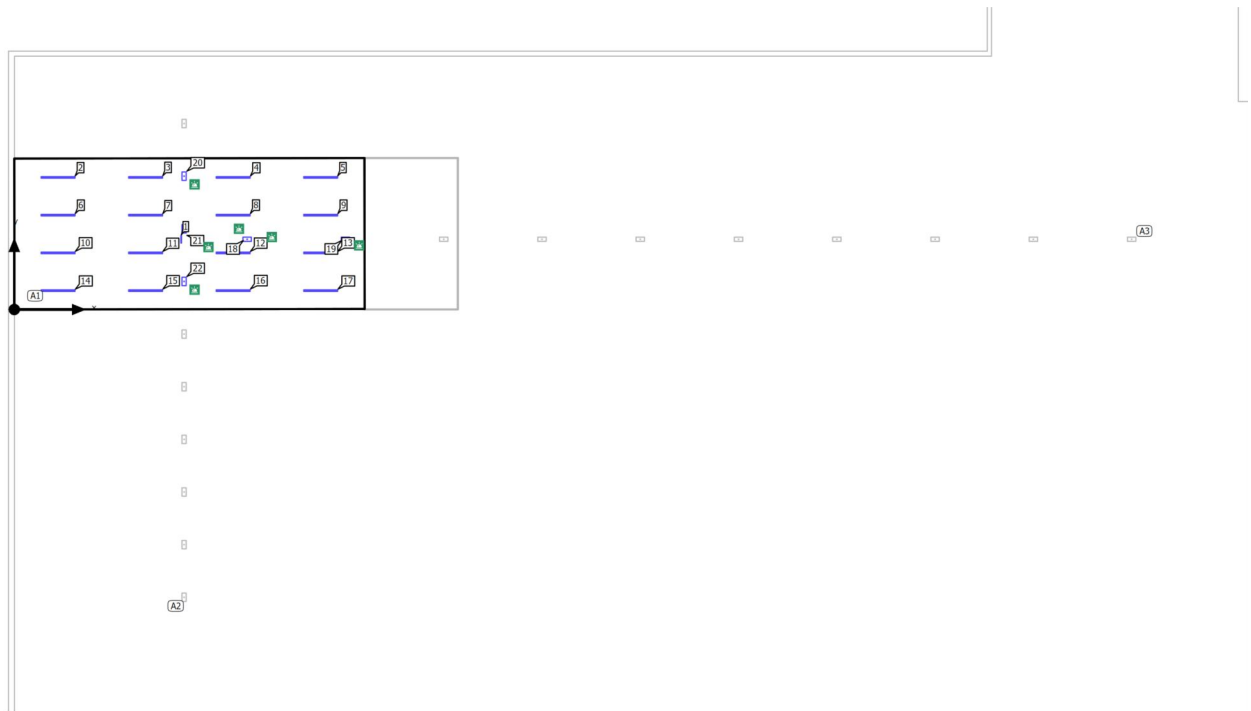
Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Luminaire list

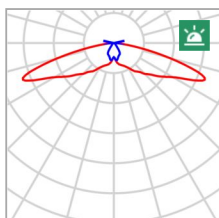
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
16	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	30	64.0 W	10782 lm	168.5 lm/W

Carpark · GROUND FLOOR · Entrance during daytime

Luminaire layout plan



Carpark · GROUND FLOOR · Entrance during daytime

Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K212/3-A-D800A40 (_LTRevAC)	Φ _{Luminaire}	275 lm
Fitting	1x light	Φ _{Emergency lighting}	275 lm
		ELF	100 %

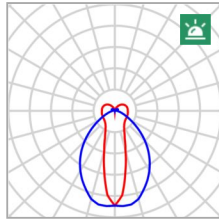
10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	7.273 m / 5.711 m / 2.800 m	7.273 m	5.711 m	2.800 m	20
X-direction	10 pcs., Centre - centre, 2.255 m	7.273 m	3.456 m	2.800 m	21
Arrangement	A2	7.273 m	1.201 m	2.800 m	22

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	9.975 m / 3.009 m / 2.800 m	9.975 m	3.009 m	2.800 m	18
X-direction	10 pcs., Centre - centre, 4.210 m	14.185 m	3.009 m	2.800 m	19
Arrangement	A3				

Carpark · GROUND FLOOR · Entrance during daytime

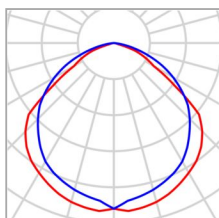
Luminaire layout plan

Manufacturer	ETAP	P	2.5 W
Article No.	K9R142/1-AX1	P _{Emergency lighting}	0.0 W
Fitting	1x sign with picto	Φ _{Luminaire}	80 lm
		Φ _{Emergency lighting}	1 lm
		ELF	1 %

Individual luminaires

X	Y	Mounting height	Luminaire
7.150 m	3.009 m	2.800 m	1

Carpark · GROUND FLOOR · Entrance during daytime

Luminaire layout plan

Manufacturer	Lena Lighting	P	64.0 W
Article No.	552489	$\Phi_{\text{Luminaire}}$	10782 lm
Article name	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W		
Fitting	1x LED GO 62W		

16 x Lena Lighting INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.875 m / 0.808 m / 2.800 m	1.875 m	5.657 m	2.800 m	2
X-direction	4 pcs., Centre - centre, 3.750 m	5.625 m	5.657 m	2.800 m	3
Y-direction	4 pcs., Centre - centre, 1.616 m	9.375 m	5.657 m	2.800 m	4
Arrangement	A1	13.125 m	5.657 m	2.800 m	5
		1.875 m	4.041 m	2.800 m	6
		5.625 m	4.041 m	2.800 m	7
		9.375 m	4.041 m	2.800 m	8
		13.125 m	4.041 m	2.800 m	9
		1.875 m	2.425 m	2.800 m	10
		5.625 m	2.425 m	2.800 m	11
		9.375 m	2.425 m	2.800 m	12
		13.125 m	2.425 m	2.800 m	13
		1.875 m	0.808 m	2.800 m	14

Carpark · GROUND FLOOR · Entrance during daytime

Luminaire layout plan

X	Y	Mounting height	Luminaire
5.625 m	0.808 m	2.800 m	15
9.375 m	0.808 m	2.800 m	16
13.125 m	0.808 m	2.800 m	17

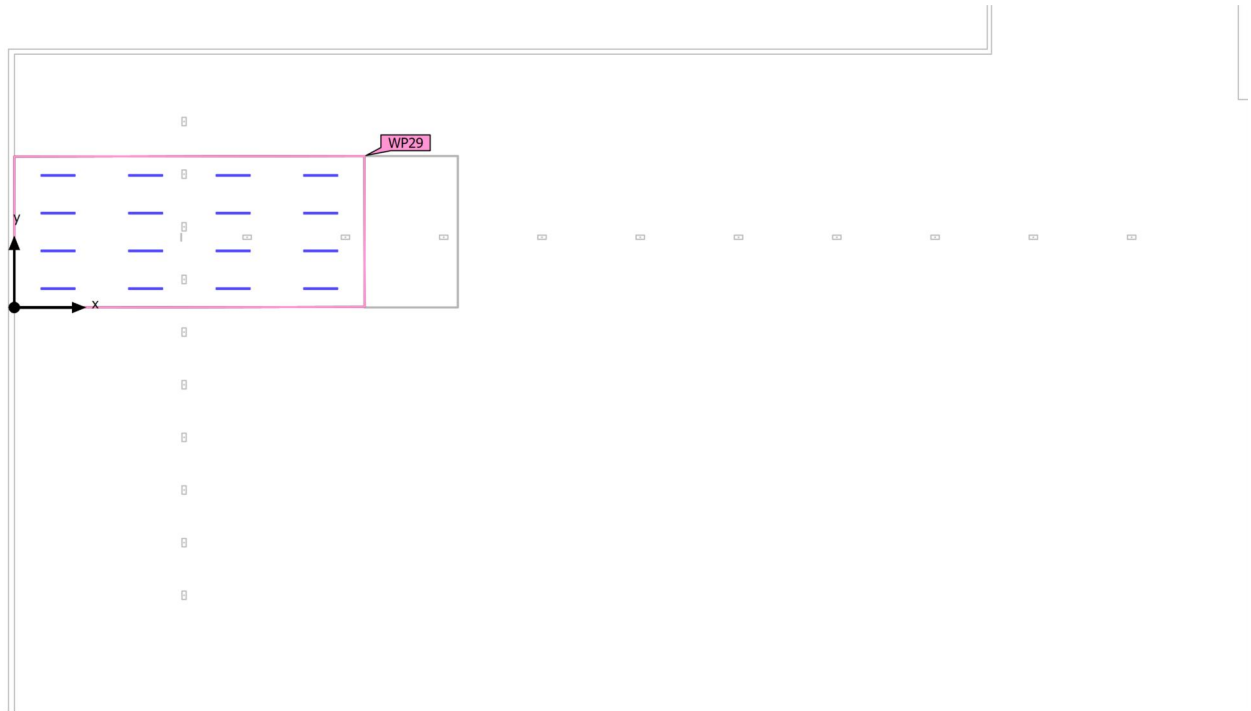
Carpark · GROUND FLOOR · Entrance during daytime

Luminaire list

Φ _{total} 173967 lm		P _{total} 1031.0 W		Luminous efficacy 168.7 lm/W		Φ _{Emergency lighting} 1376 lm		P _{Emergency lighting} 4.5 W	
pcs.	Manufacturer	Article No.	Article name				P	Φ	Luminous efficacy
5	ETAP	-	K212/3-A-D800A40	(_LTRevAC)		0.9 W	275 lm	312.5 lm/W	
						0.9 W	275 lm (100 %)	–	
1	ETAP	K9R142/1-AX1				2.5 W	80 lm	32.0 lm/W	
						0.0 W	1 lm (1 %)	–	
16	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W				64.0 W	10782 lm	168.5 lm/W

Carpark · GROUND FLOOR · Entrance during daytime (Light scene 1)

Calculation objects



Carpark · GROUND FLOOR · Entrance during daytime (Light scene 1)

Calculation objects

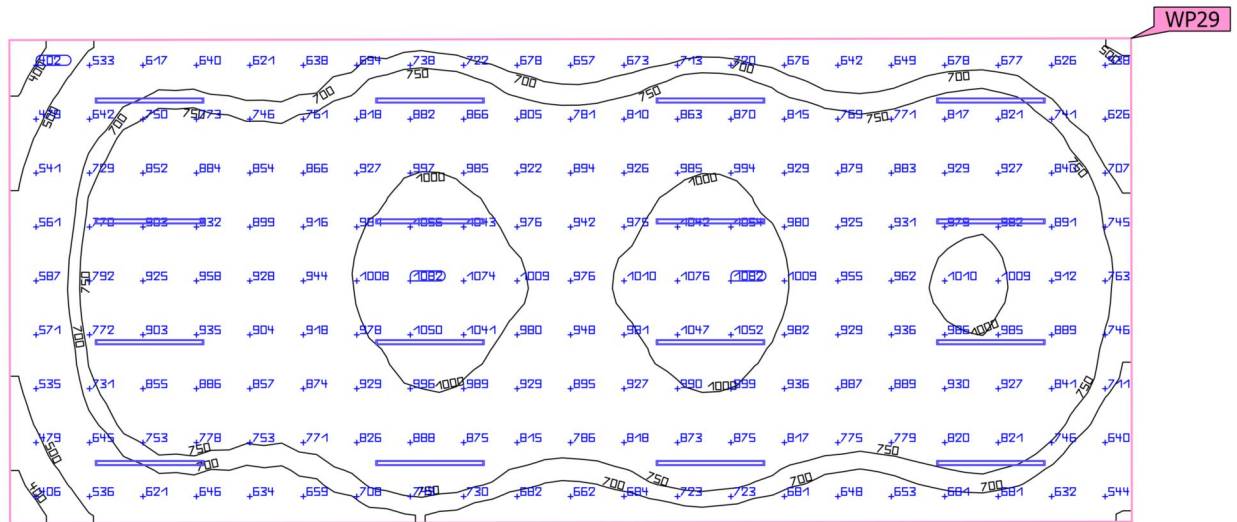
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Entrance during daytime) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	825 lx (≥ 800 lx) ✓	345 lx	1090 lx	0.42 (≥ 0.40) ✓	0.32	WP29

Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Carpark · GROUND FLOOR · Entrance during daytime (Light scene 1)

Working plane (Entrance during daytime)



Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Entrance during daytime)	825 lx	345 lx	1090 lx	0.42	0.32	WP29
Perpendicular illuminance (adaptive)	(≥ 800 lx)			(≥ 0.40)		
Height: 0.000 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m (Light scene 1)

Summary



Ground area	25.95 m²		
Reflection factors	Ceiling: 70.0 %, Walls: 0.0 %, Floor: 20.0 %	Mounting height	2.800 m
		Height _{Working plane}	0.000 m
Maintenance factor	0.62 (fixed)	Wall zone _{Working plane}	0.000 m

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	459 lx	$\geq 400 \text{ lx}$	✓	WP30
	$U_o (g_1)$	0.44	≥ 0.40	✓	WP30
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	31	≤ 25	✗	
Energy estimation ⁽²⁾	Consumption	141 kWh/a	max. 950 kWh/a	✓	
Space	Lighting power density	4.93 W/m ²	–		
		1.07 W/m ² /100 lx	–		

(1) Based on a rectangular space of 6.487 m x 4.000 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

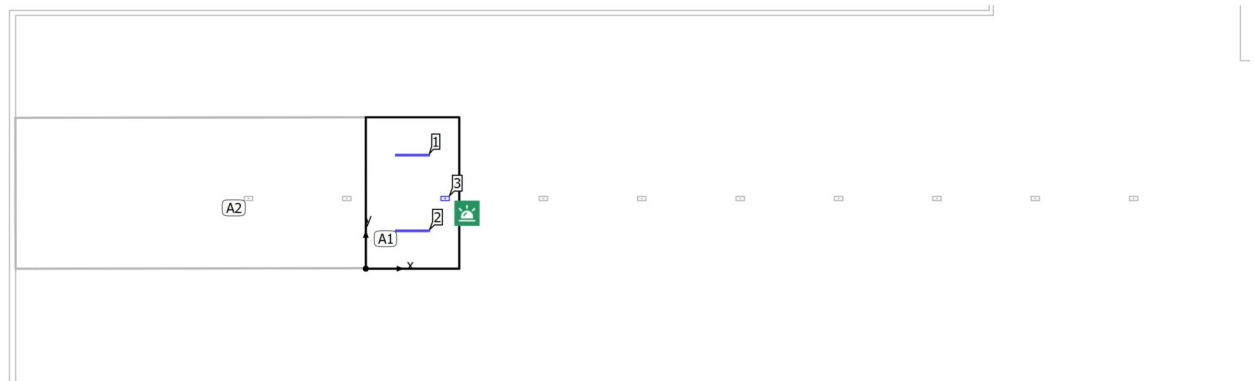
Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Luminaire list

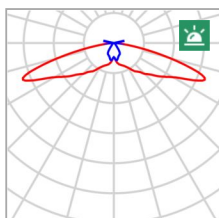
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
2	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W	31	64.0 W	10782 lm	168.5 lm/W

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m

Luminaire layout plan



Carpark · GROUND FLOOR · Entrance during daytime — next 4 m

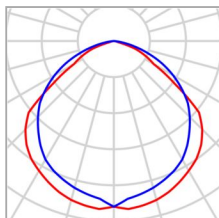
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K212/3-A-D800A40 (_LTRevAC)	Φ _{Luminaire}	275 lm
Fitting	1x light	Φ _{Emergency lighting}	275 lm
		ELF	100 %

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	3.395 m / 3.007 m / 2.800 m	3.395 m	3.007 m	2.800 m	3
X-direction	10 pcs., Centre - centre, 4.210 m				
Arrangement	A2				

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m

Luminaire layout plan

Manufacturer	Lena Lighting	P	64.0 W
Article No.	552489	$\Phi_{\text{Luminaire}}$	10782 lm
Article name	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W		
Fitting	1x LED GO 62W		

2 x Lena Lighting INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	2.000 m / 1.622 m / 2.800 m	2.000 m	4.865 m	2.800 m	1
X-direction	1 pcs., Centre - centre, 4.000 m	2.000 m	1.622 m	2.800 m	2
Y-direction	2 pcs., Centre - centre, 3.243 m				
Arrangement	A1				

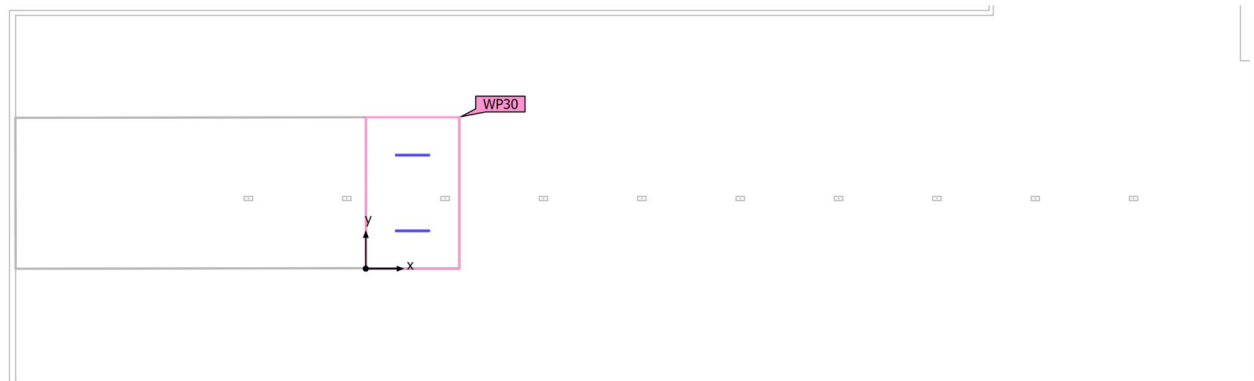
Carpark · GROUND FLOOR · Entrance during daytime — next 4 m

Luminaire list

Φ_{total}		P_{total}		Luminous efficacy		$\Phi_{\text{Emergency lighting}}$		$P_{\text{Emergency lighting}}$	
21839 lm		128.9 W		169.4 lm/W		275 lm		0.9 W	
pcs.	Manufacturer	Article No.	Article name				P	Φ	Luminous efficacy
1	ETAP	-	K212/3-A-D800A40 (_LTRevAC)				0.9 W	275 lm	312.5 lm/W
							 0.9 W	275 lm (100 %)	-
2	Lena Lighting	552489	INDUSTRY LED ENDURA 10500lm 1450mm 840 IP66 62W				64.0 W	10782 lm	168.5 lm/W

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m (Light scene 1)

Calculation objects



Carpark · GROUND FLOOR · Entrance during daytime — next 4 m (Light scene 1)

Calculation objects

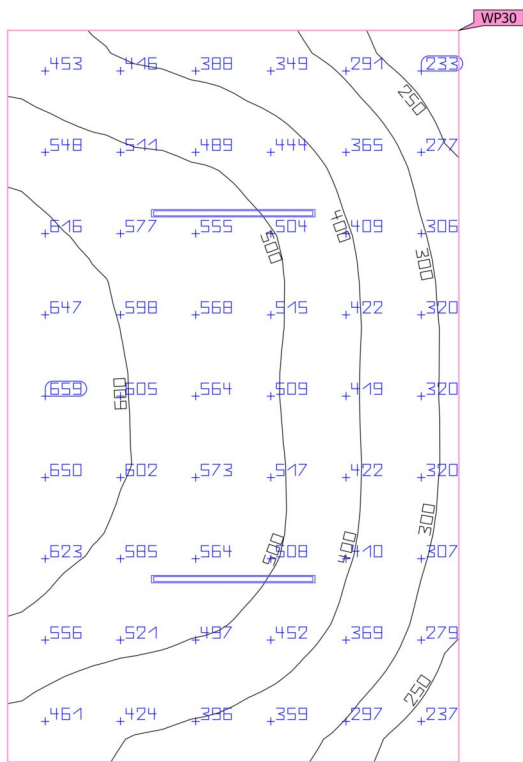
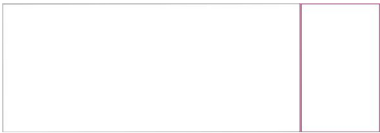
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Entrance during daytime — next 4 m) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	459 lx (≥ 400 lx) ✓	201 lx	686 lx	0.44 (≥ 0.40) ✓	0.29	WP30

Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Carpark · GROUND FLOOR · Entrance during daytime — next 4 m (Light scene 1)

Working plane (Entrance during daytime — next 4 m)

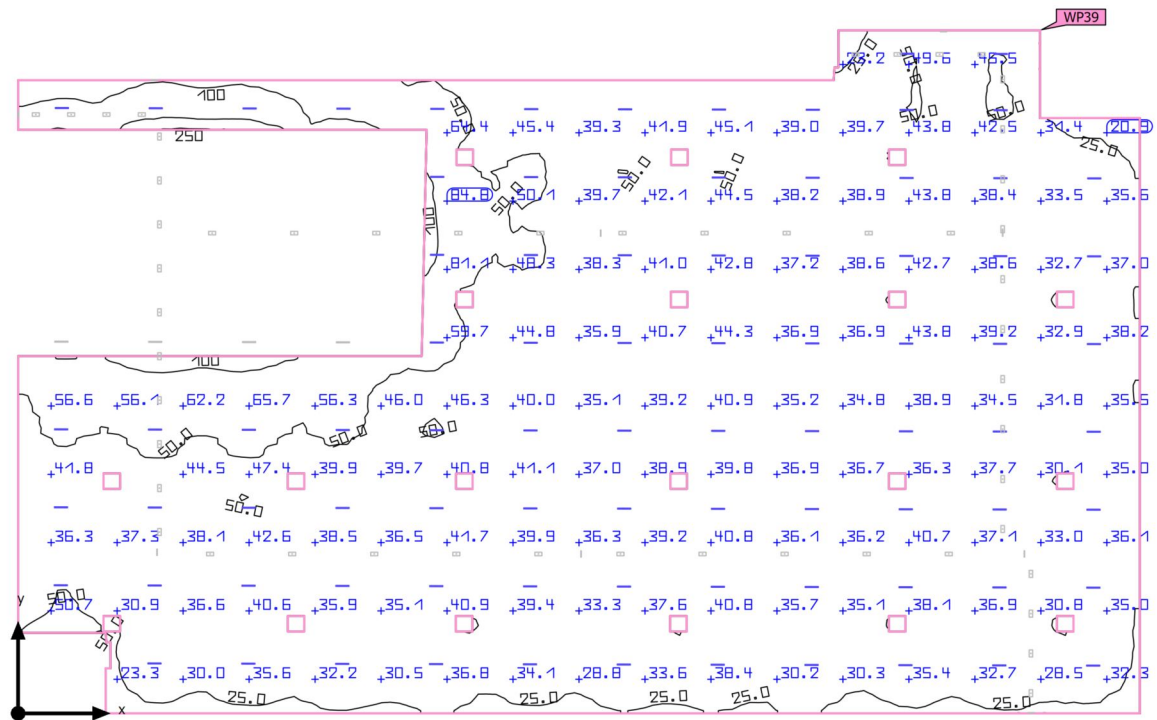


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Entrance during daytime — next 4 m)	459 lx	201 lx	686 lx	0.44	0.29	WP30
Perpendicular illuminance (adaptive)	(≥ 400 lx)			(≥ 0.40)		
Height: 0.000 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: Traffic zones inside buildings (9.4 Area in front of lift, escalators)

Carpark · GROUND FLOOR · Exclude carpark (Light scene 1)

Summary



Ground area	1625.11 m ²		
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.54 (fixed)	Height _{Working plane}	0.000 m
		Wall zone _{Working plane}	0.000 m

Carpark · GROUND FLOOR · Exclude carpark (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	43.0 lx	$\geq 40.0 \text{ lx}$	✓	WP39
	$U_o (g_1)$	0.37	≥ 0.30	✓	WP39
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	24	≤ 28	✓	
Energy estimation ⁽²⁾	Consumption	1377 kWh/a	max. 56900 kWh/a	✓	
Space	Lighting power density	0.77 W/m ²	–		
		1.79 W/m ² /100 lx	–		

(1) Based on a rectangular space of 57.535 m x 35.050 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

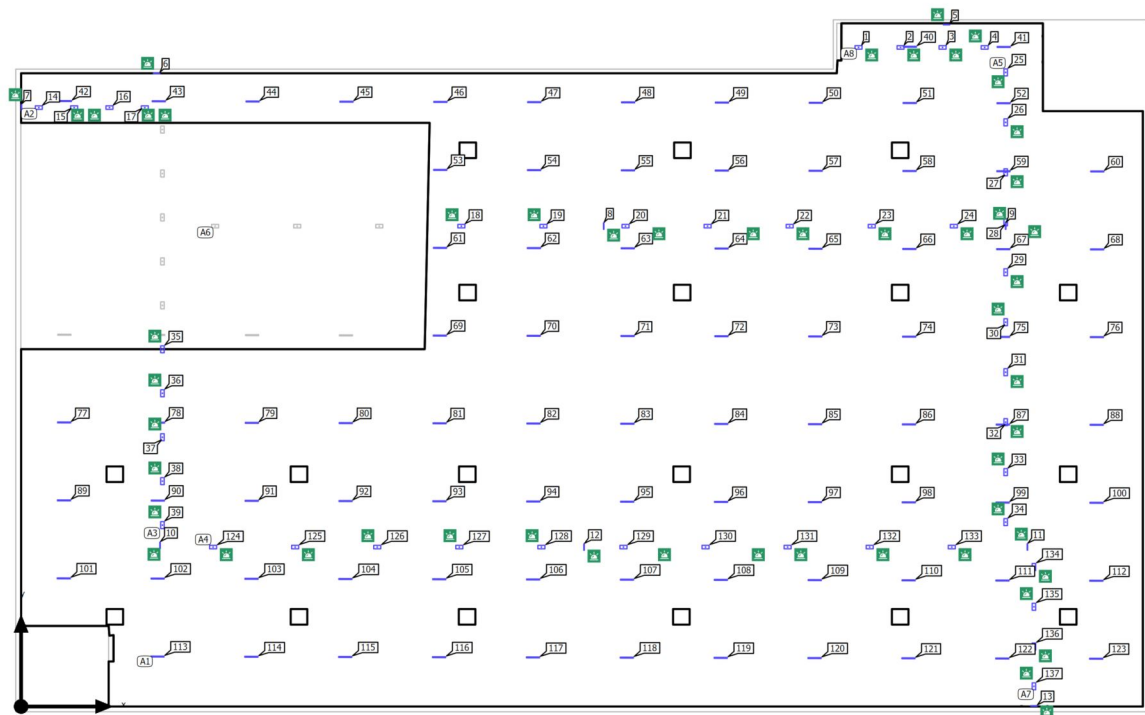
Utilisation profile: Traffic zones inside buildings (9.1 Circulation areas and corridors)

Luminaire list

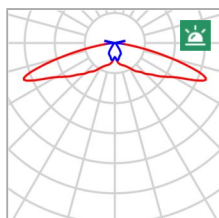
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
84	Lena Lighting	100765	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)	24	14.9 W	1650 lm	110.7 lm/W

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan



Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K212/3-A-D800A40 (_LTRevAC)	Φ _{Luminaire}	275 lm
Fitting	1x light	Φ _{Emergency lighting}	275 lm
		ELF	100 %

4 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.906 m / 30.728 m / 2.800 m	0.906 m	30.728 m	2.800 m	14
X-direction	4 pcs., Centre - centre, 1.812 m	2.718 m	30.728 m	2.800 m	15
		4.530 m	30.728 m	2.800 m	16
Arrangement	A2	6.342 m	30.728 m	2.800 m	17

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	7.248 m / 18.326 m / 2.800 m	7.248 m	18.326 m	2.800 m	35
X-direction	10 pcs., Centre - centre, 2.255 m	7.248 m	16.071 m	2.800 m	36
		7.248 m	13.816 m	2.800 m	37
Arrangement	A3	7.248 m	11.561 m	2.800 m	38
		7.248 m	9.306 m	2.800 m	39

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	9.845 m / 8.178 m / 2.800 m	9.845 m	8.178 m	2.800 m	124
X-direction	10 pcs., Centre - centre, 4.210 m	14.055 m	8.178 m	2.800 m	125
Arrangement	A4	18.265 m	8.178 m	2.800 m	126
		22.475 m	8.178 m	2.800 m	127
		26.685 m	8.178 m	2.800 m	128
		30.895 m	8.178 m	2.800 m	129
		35.105 m	8.178 m	2.800 m	130
		39.315 m	8.178 m	2.800 m	131
		43.525 m	8.178 m	2.800 m	132
		47.735 m	8.178 m	2.800 m	133

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	50.498 m / 32.532 m / 2.800 m	50.498 m	32.532 m	2.800 m	25
X-direction	10 pcs., Centre - centre, 2.564 m	50.498 m	29.969 m	2.800 m	26
Arrangement	A5	50.498 m	27.405 m	2.800 m	27
		50.498 m	24.842 m	2.800 m	28
		50.498 m	22.278 m	2.800 m	29
		50.498 m	19.714 m	2.800 m	30
		50.498 m	17.151 m	2.800 m	31
		50.498 m	14.587 m	2.800 m	32
		50.498 m	12.024 m	2.800 m	33

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

X	Y	Mounting height	Luminaire
50.498 m	9.460 m	2.800 m	34

10 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	22.580 m / 24.643 m / 2.800 m	22.580 m	24.643 m	2.800 m	18
X-direction	10 pcs., Centre - centre, 4.210 m	26.790 m	24.643 m	2.800 m	19
		31.000 m	24.643 m	2.800 m	20
Arrangement	A6	35.210 m	24.643 m	2.800 m	21
		39.420 m	24.643 m	2.800 m	22
		43.630 m	24.643 m	2.800 m	23
		47.840 m	24.643 m	2.800 m	24

4 x ETAP K212/3-A-D800A40 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	51.948 m / 7.158 m / 2.800 m	51.948 m	7.158 m	2.800 m	134
X-direction	4 pcs., Centre - centre, 2.041 m	51.948 m	5.117 m	2.800 m	135
		51.948 m	3.076 m	2.800 m	136
Arrangement	A7	51.948 m	1.036 m	2.800 m	137

4 x ETAP K212/3-A-D800A40 (_LTRevAC)

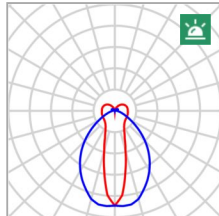
Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	42.951 m / 33.814 m / 2.800 m	42.951 m	33.814 m	2.800 m	1
		45.107 m	33.814 m	2.800 m	2

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

X-direction	4 pcs., Centre - centre, 2.156 m	X	Y	Mounting height	Luminaire
Arrangement	A8	47.263 m	33.814 m	2.800 m	3
		49.420 m	33.814 m	2.800 m	4

Carpark · GROUND FLOOR · Exclude carpark

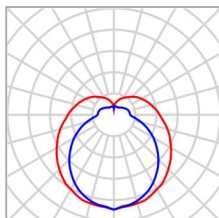
Luminaire layout plan

Manufacturer	ETAP	P	2.5 W
Article No.	K9R142/1-AX1	P _{Emergency lighting}	0.0 W
Fitting	1x sign with picto	Φ _{Luminaire}	80 lm
		Φ _{Emergency lighting}	1 lm
		ELF	1 %

Individual luminaires

X	Y	Mounting height	Luminaire
47.465 m	34.984 m	2.800 m	5
6.947 m	32.487 m	2.800 m	6
0.013 m	30.728 m	2.800 m	7
29.880 m	24.643 m	2.800 m	8
50.498 m	24.643 m	2.800 m	9
7.125 m	8.268 m	2.800 m	10
51.612 m	8.184 m	2.800 m	11
28.873 m	8.178 m	2.800 m	12
51.948 m	0.030 m	2.800 m	13

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

Manufacturer	Lena Lighting	P	14.9 W
Article No.	100765	Φ _{Luminaire}	1650 lm
Article name	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)		
Fitting	1x LED GO 13W		

88 x Lena Lighting MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	2.188 m / 6.573 m / 2.800 m	45.583 m	33.848 m	2.800 m	40
X-direction	12 pcs., Outer edge - outer edge, Distances not equal	50.398 m	33.837 m	2.800 m	41
		2.244 m	31.062 m	2.800 m	42
Y-direction	9 pcs., Centre - centre, Distances not equal	7.059 m	31.051 m	2.800 m	43
		11.874 m	31.040 m	2.800 m	44
Arrangement	A1	16.688 m	31.029 m	2.800 m	45
		21.503 m	31.018 m	2.800 m	46
		26.318 m	31.006 m	2.800 m	47
		31.132 m	30.995 m	2.800 m	48
		35.947 m	30.984 m	2.800 m	49
		40.762 m	30.973 m	2.800 m	50
		45.576 m	30.962 m	2.800 m	51
		50.391 m	30.951 m	2.800 m	52

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

X	Y	Mounting height	Luminaire
21.495 m	27.529 m	2.800 m	53
26.309 m	27.518 m	2.800 m	54
31.124 m	27.507 m	2.800 m	55
35.939 m	27.495 m	2.800 m	56
40.753 m	27.484 m	2.800 m	57
45.568 m	27.473 m	2.800 m	58
50.383 m	27.462 m	2.800 m	59
55.197 m	27.451 m	2.800 m	60
21.486 m	23.529 m	2.800 m	61
26.300 m	23.518 m	2.800 m	62
31.115 m	23.507 m	2.800 m	63
35.930 m	23.495 m	2.800 m	64
40.744 m	23.484 m	2.800 m	65
45.559 m	23.473 m	2.800 m	66
50.374 m	23.462 m	2.800 m	67
55.188 m	23.451 m	2.800 m	68
21.475 m	19.029 m	2.800 m	69
26.290 m	19.018 m	2.800 m	70
31.104 m	19.007 m	2.800 m	71
35.919 m	18.996 m	2.800 m	72
40.734 m	18.984 m	2.800 m	73
45.548 m	18.973 m	2.800 m	74
50.363 m	18.962 m	2.800 m	75
55.178 m	18.951 m	2.800 m	76

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

X	Y	Mounting height	Luminaire
2.206 m	14.573 m	2.800 m	77
7.021 m	14.562 m	2.800 m	78
11.835 m	14.551 m	2.800 m	79
16.650 m	14.540 m	2.800 m	80
21.465 m	14.529 m	2.800 m	81
26.279 m	14.518 m	2.800 m	82
31.094 m	14.507 m	2.800 m	83
35.909 m	14.496 m	2.800 m	84
40.723 m	14.484 m	2.800 m	85
45.538 m	14.473 m	2.800 m	86
50.353 m	14.462 m	2.800 m	87
55.167 m	14.451 m	2.800 m	88
2.197 m	10.573 m	2.800 m	89
7.011 m	10.562 m	2.800 m	90
11.826 m	10.551 m	2.800 m	91
16.641 m	10.540 m	2.800 m	92
21.455 m	10.529 m	2.800 m	93
26.270 m	10.518 m	2.800 m	94
31.085 m	10.507 m	2.800 m	95
35.899 m	10.496 m	2.800 m	96
40.714 m	10.484 m	2.800 m	97
45.529 m	10.473 m	2.800 m	98
50.343 m	10.462 m	2.800 m	99
55.158 m	10.451 m	2.800 m	100

Carpark · GROUND FLOOR · Exclude carpark

Luminaire layout plan

X	Y	Mounting height	Luminaire
2.188 m	6.573 m	2.800 m	101
7.002 m	6.562 m	2.800 m	102
11.817 m	6.551 m	2.800 m	103
16.632 m	6.540 m	2.800 m	104
21.446 m	6.529 m	2.800 m	105
26.261 m	6.518 m	2.800 m	106
31.076 m	6.507 m	2.800 m	107
35.890 m	6.496 m	2.800 m	108
40.705 m	6.484 m	2.800 m	109
45.520 m	6.473 m	2.800 m	110
50.334 m	6.462 m	2.800 m	111
55.149 m	6.451 m	2.800 m	112
6.993 m	2.562 m	2.800 m	113
11.808 m	2.551 m	2.800 m	114
16.622 m	2.540 m	2.800 m	115
21.437 m	2.529 m	2.800 m	116
26.252 m	2.518 m	2.800 m	117
31.066 m	2.507 m	2.800 m	118
35.881 m	2.496 m	2.800 m	119
40.696 m	2.484 m	2.800 m	120
45.510 m	2.473 m	2.800 m	121
50.325 m	2.462 m	2.800 m	122
55.140 m	2.451 m	2.800 m	123

Carpark · GROUND FLOOR · Exclude carpark

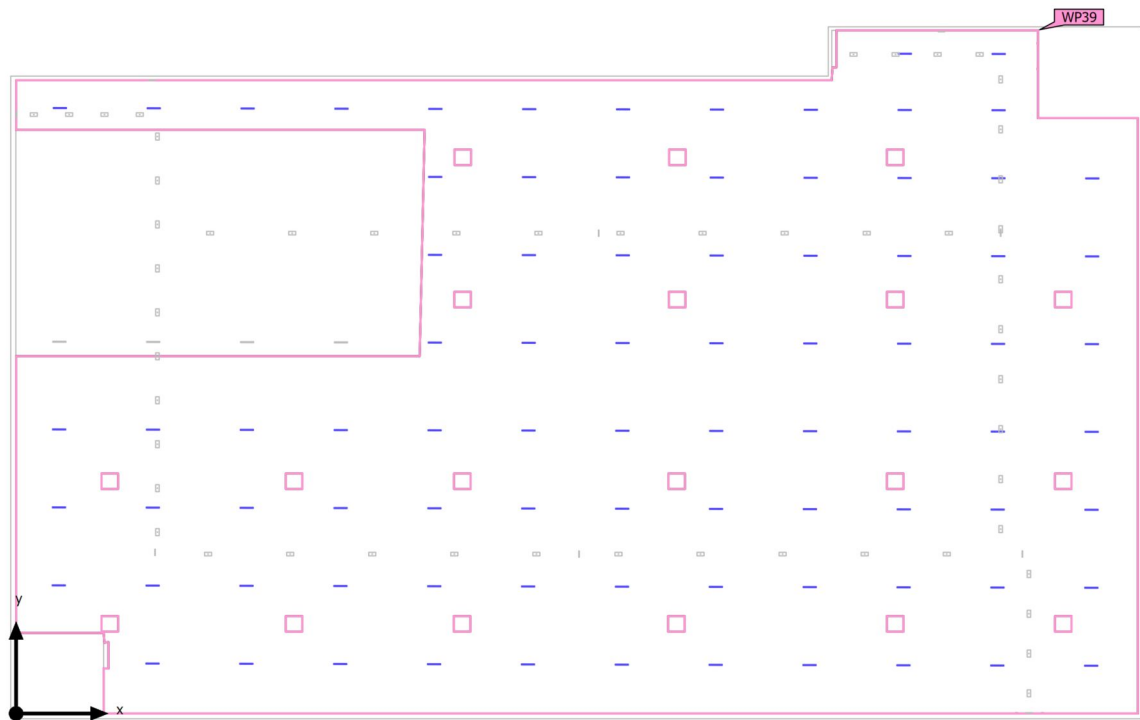
Luminaire list

Φ_{total} 151420 lm	P_{total} 1313.7 W	Luminous efficacy 115.3 lm/W	$\Phi_{\text{Emergency lighting}}$ 12109 lm	$P_{\text{Emergency lighting}}$ 39.6 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
44	ETAP	-	K212/3-A-D800A40 (_LTRRevAC)	0.9 W	275 lm	312.5 lm/W
				 0.9 W	275 lm (100 %)	-
9	ETAP	K9R142/1-AX1		2.5 W	80 lm	32.0 lm/W
				 0.0 W	1 lm (1 %)	-
84	Lena Lighting	100765	MIMO 2 LED BASIC 670mm 1650lm IP66 840 (13W)	14.9 W	1650 lm	110.7 lm/W

Carpark · GROUND FLOOR · Exclude carpark (Light scene 1)

Calculation objects



Carpark · GROUND FLOOR · Exclude carpark (Light scene 1)

Calculation objects

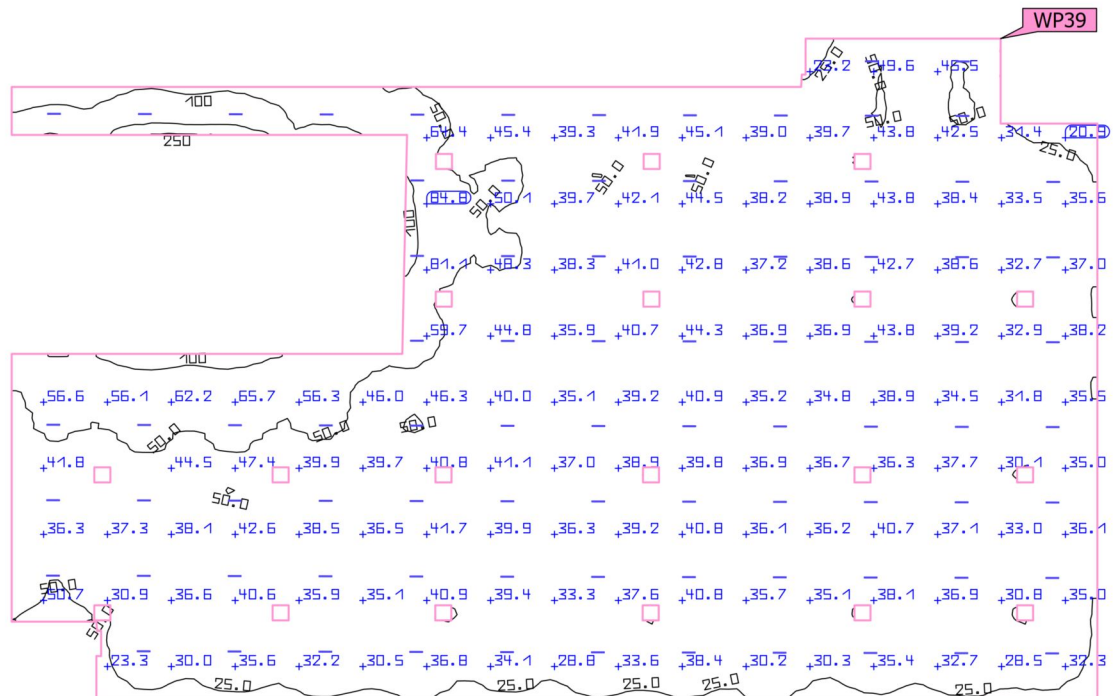
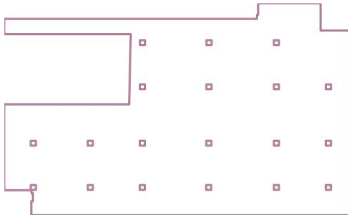
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Exclude carpark) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	43.0 lx (≥ 40.0 lx) ✓	15.9 lx	254 lx	0.37 (≥ 0.30) ✓	0.063	WP39

Utilisation profile: Traffic zones inside buildings (9.1 Circulation areas and corridors)

Carpark · GROUND FLOOR · Exclude carpark (Light scene 1)

Working plane (Exclude carpark)

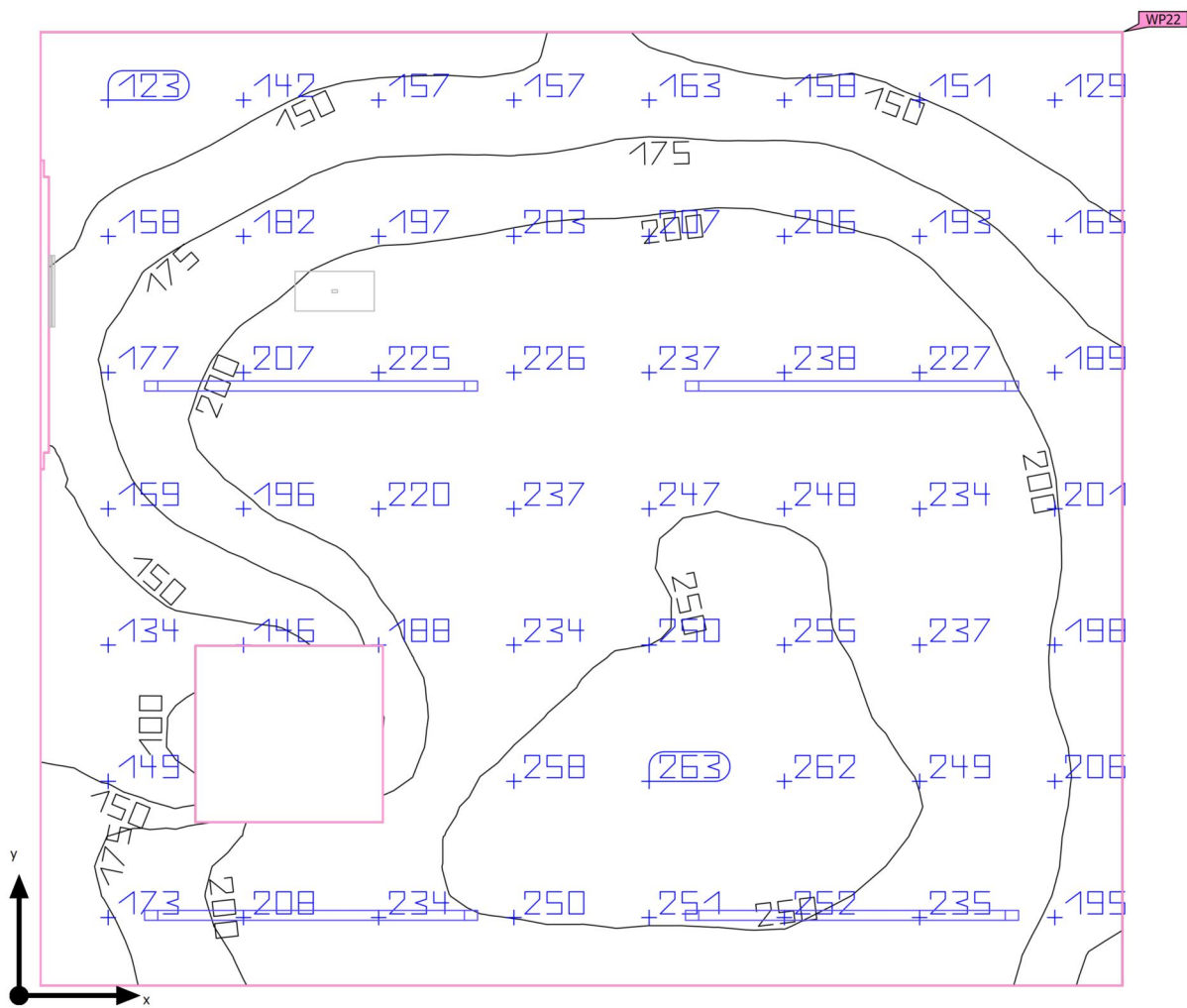


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_0 (g_1)$ (Target)	g_2	Index
Working plane (Exclude carpark)	43.0 lx	15.9 lx	254 lx	0.37	0.063	WP39
Perpendicular illuminance (adaptive)	(≥ 40.0 lx)			(≥ 0.30)		
Height: 0.000 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: Traffic zones inside buildings (9.1 Circulation areas and corridors)

Carpark · GROUND FLOOR · GENERATOR (Light scene 1)

Summary



Ground area	21.21 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.54 (fixed)

Clearance height	2.800 m
Mounting height	2.800 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Carpark · GROUND FLOOR · GENERATOR (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	203 lx	≥ 160 lx	✓	WP22
	$U_o (g_1)$	0.45	≥ 0.40	✓	WP22
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	21	≤ 25	✓	
Energy estimation ⁽²⁾	Consumption	385 kWh/a	max. 750 kWh/a	✓	
Space	Lighting power density	7.34 W/m ²	–		
		3.61 W/m ² /100 lx	–		

(1) Based on a rectangular space of 4.324 m x 4.905 m and SHR of 0.25.

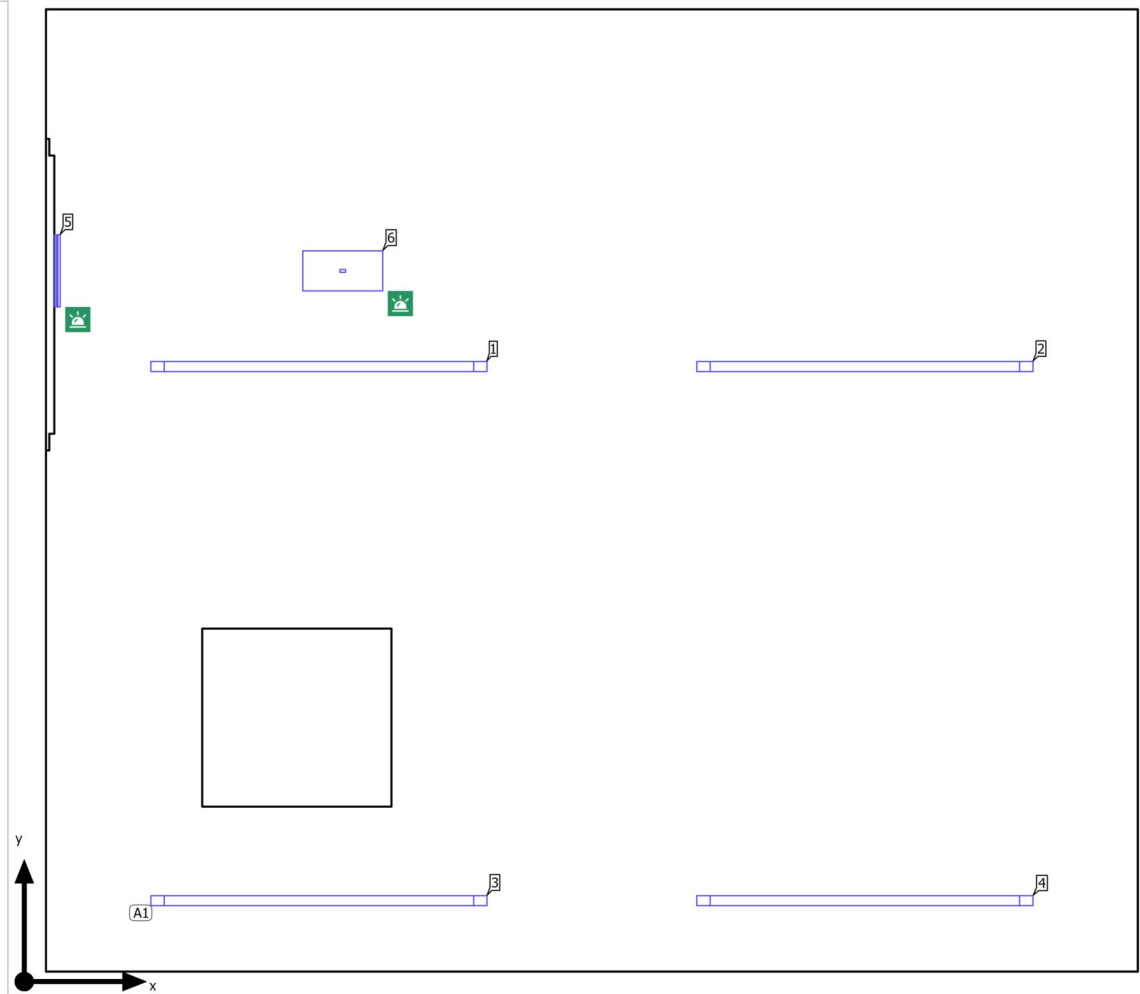
(2) Calculated using DIN:18599-4.

Utilisation profile: General areas inside buildings - Store rooms, cold stores (12.1 Store and stockrooms)

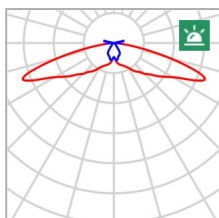
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	21	38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR · GENERATOR

Luminaire layout plan

Carpark · GROUND FLOOR · GENERATOR

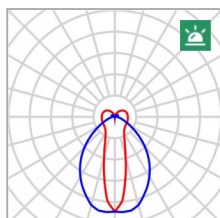
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K212/3-A-D800A40 (_LTRevAC)	Φ _{Luminaire}	275 lm
Fitting	1x light	Φ _{Emergency lighting}	275 lm
		ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.431 m	3.194 m	2.800 m	6

Carpark · GROUND FLOOR · GENERATOR

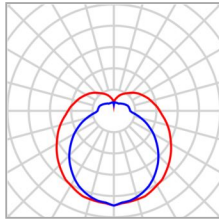
Luminaire layout plan

Manufacturer	ETAP	P	2.5 W
Article No.	K9R142/1-AX1	P _{Emergency lighting}	0.0 W
Fitting	1x sign with picto	Φ _{Luminaire}	80 lm
		Φ _{Emergency lighting}	1 lm
		ELF	1 %

Individual luminaires

X	Y	Mounting height	Luminaire
0.148 m	3.194 m	2.800 m	5

Carpark · GROUND FLOOR · GENERATOR

Luminaire layout plan

Manufacturer	Lena Lighting	P	38.9 W
Article No.	100857	$\Phi_{\text{Luminaire}}$	4550 lm
Article name	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)		
Fitting	1x LED GO 33W		

4 x Lena Lighting MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.324 m / 0.364 m / 2.800 m	1.324 m	2.764 m	2.800 m	1
X-direction	2 pcs., Centre - centre, Distances not equal	3.776 m	2.764 m	2.800 m	2
		1.324 m	0.364 m	2.800 m	3
Y-direction	2 pcs., Centre - centre, Distances not equal	3.776 m	0.364 m	2.800 m	4
Arrangement	A1				

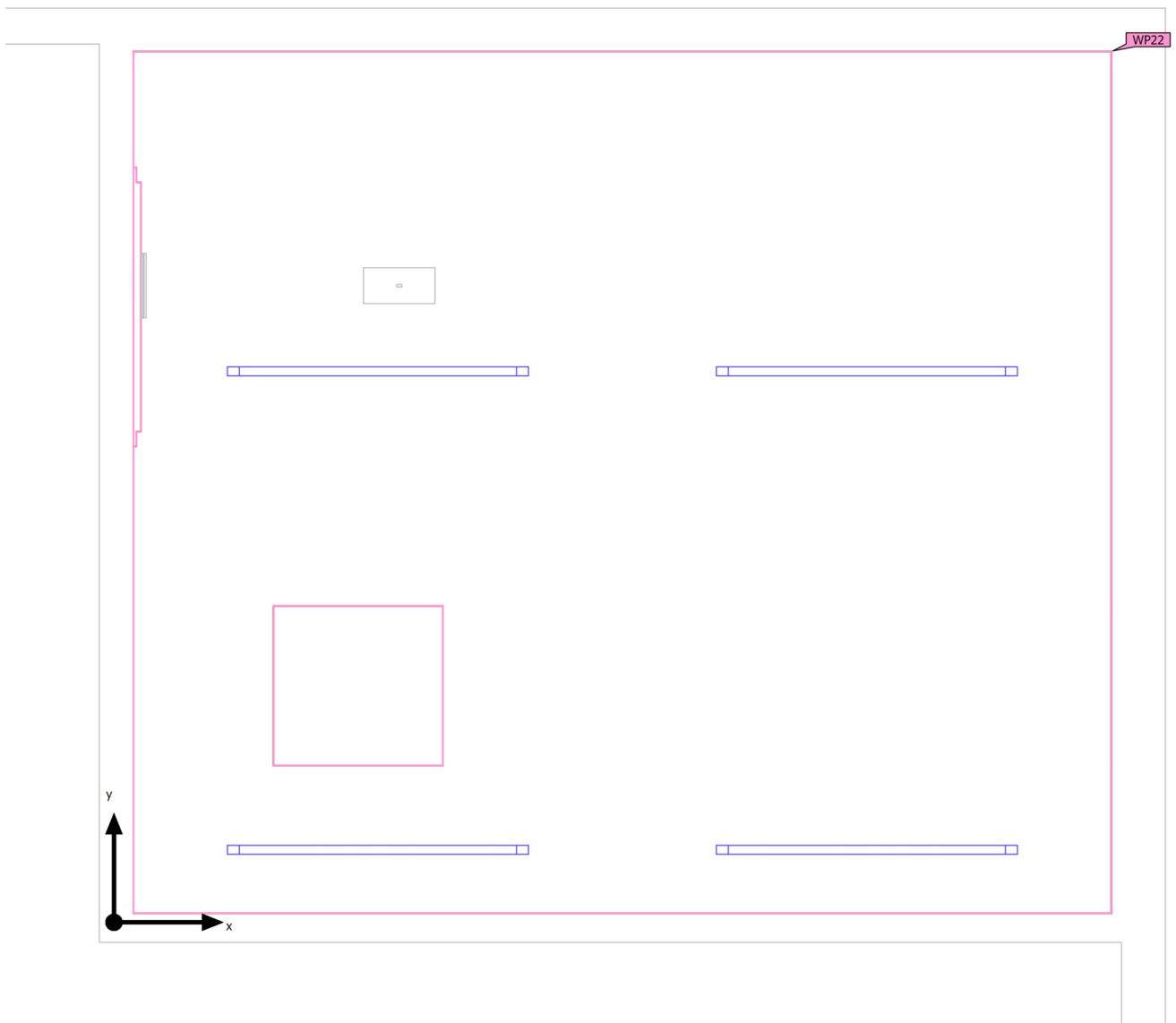
Carpark · GROUND FLOOR · GENERATOR

Luminaire list

Φ_{total} 18555 lm		P_{total} 159.0 W		Luminous efficacy 116.7 lm/W		$\Phi_{\text{Emergency lighting}}$ 276 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name				P	Φ	Luminous efficacy
1	ETAP	-	K212/3-A-D800A40 (L_TRevAC)				0.9 W	275 lm	312.5 lm/W
							 0.9 W	275 lm (100 %)	-
1	ETAP	K9R142/1-AX1					2.5 W	80 lm	32.0 lm/W
							 0.0 W	1 lm (1 %)	-
4	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)				38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR · GENERATOR (Light scene 1)

Calculation objects



Carpark · GROUND FLOOR · GENERATOR (Light scene 1)

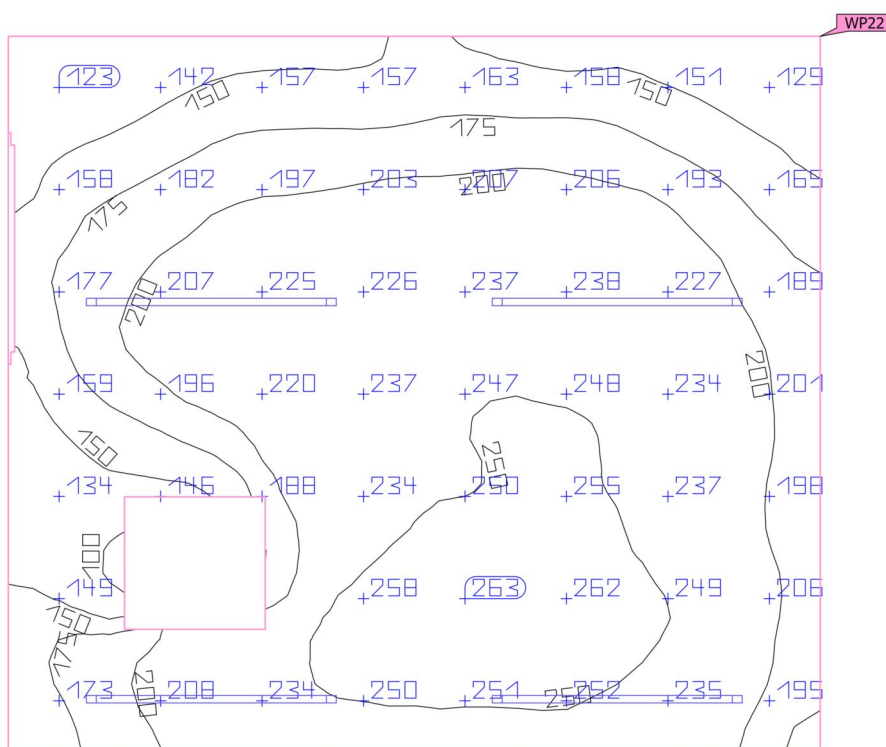
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (GENERATOR) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	203 lx (≥ 160 lx) ✓	91.6 lx	266 lx	0.45 (≥ 0.40) ✓	0.34	WP22

Utilisation profile: General areas inside buildings - Store rooms, cold stores (12.1 Store and stockrooms)

Working plane (GENERATOR)

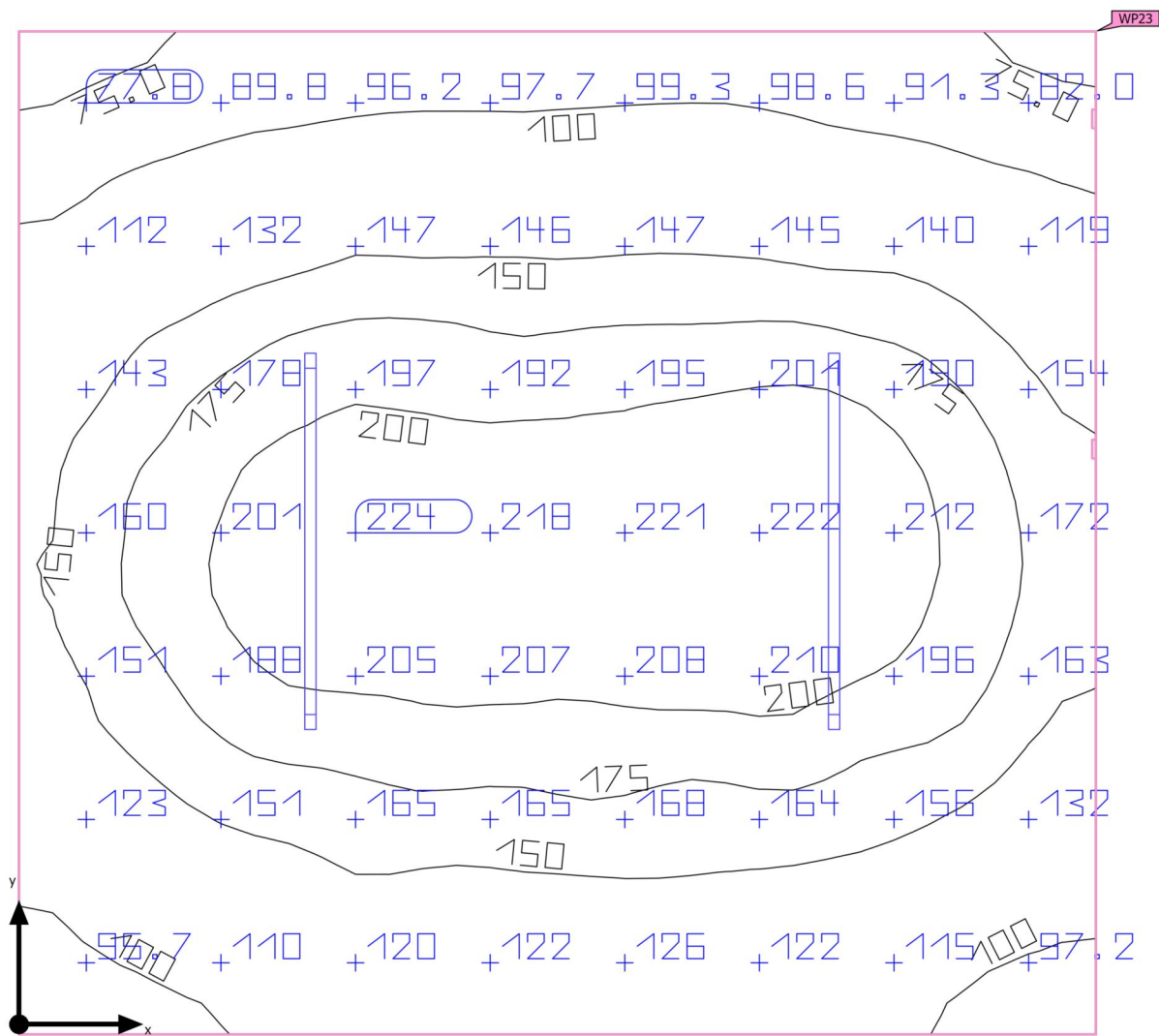


Properties	$\bar{E}$ (Target)	$E_{\min}$	$E_{\max}$	$U_o(g_1)$ (Target)	g_2	Index
Working plane (GENERATOR) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	203 lx (≥ 160 lx) ✓	91.6 lx	266 lx	0.45 (≥ 0.40) ✓	0.34	WP22

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Carpark · GROUND FLOOR · STORAGE (Light scene 1)

Summary



Ground area	17.40 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.62 (fixed)

Clearance height	2.800 m
Mounting height	2.800 m
Height _{Working plane}	0.800 m
Wall zone _{Working plane}	0.000 m

Carpark · GROUND FLOOR · STORAGE (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	153 lx	≥ 80.0 lx	✓	WP23
	$U_o (g_1)$	0.43	≥ 0.40	✓	WP23
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	20	≤ 25	✓	
Energy estimation ⁽²⁾	Consumption	193 kWh/a	max. 650 kWh/a	✓	
Space	Lighting power density	4.47 W/m ²	–		
		2.93 W/m ² /100 lx	–		

(1) Based on a rectangular space of 4.321 m x 4.026 m and SHR of 0.25.

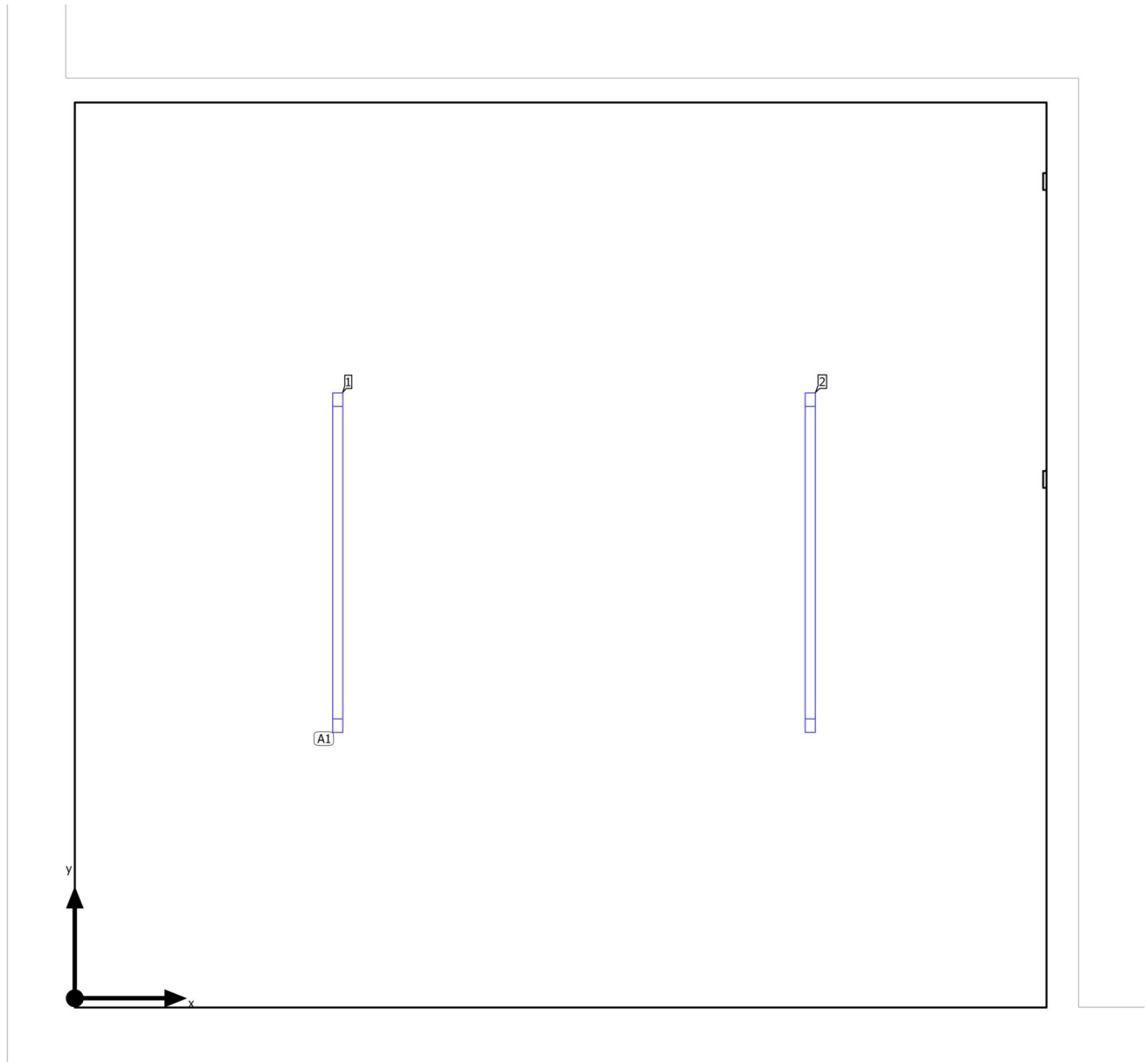
(2) Calculated using DIN:18599-4.

Utilisation profile: General areas inside buildings - Store rooms, cold stores (12.1 Store and stockrooms)

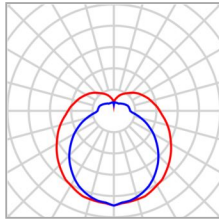
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
2	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	20	38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR · STORAGE

Luminaire layout plan

Carpark · GROUND FLOOR · STORAGE

Luminaire layout plan

Manufacturer	Lena Lighting	P	38.9 W
Article No.	100857	$\Phi_{\text{Luminaire}}$	4550 lm
Article name	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)		
Fitting	1x LED GO 33W		

2 x Lena Lighting MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.169 m / 1.938 m / 2.800 m	1.169 m	1.938 m	2.800 m	1
X-direction	2 pcs., Centre - centre, 2.101 m	3.271 m	1.938 m	2.800 m	2
Arrangement	A1				

Carpark · GROUND FLOOR · STORAGE

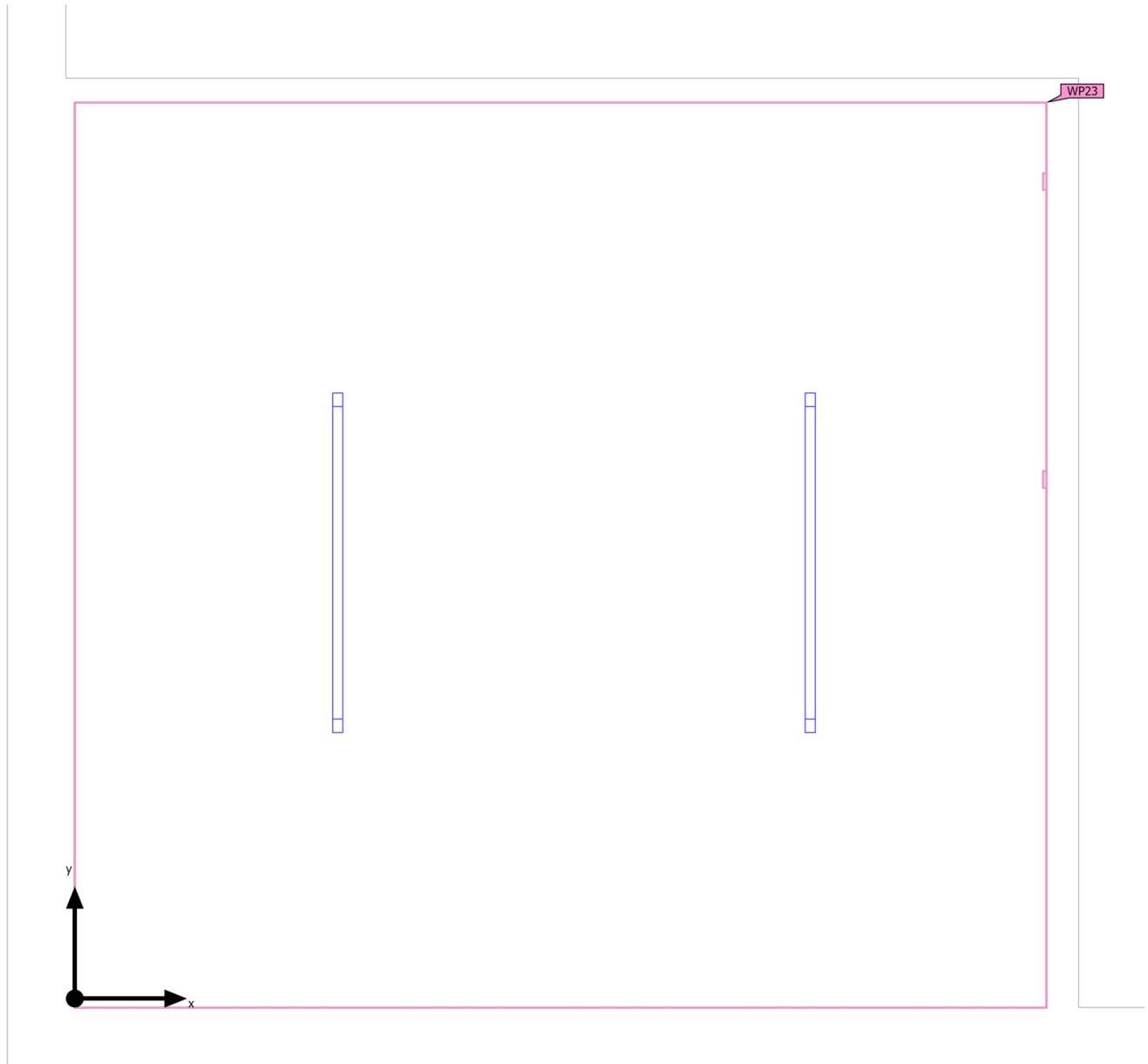
Luminaire list

Φ_{total} 9100 lm	P_{total} 77.8 W	Luminous efficacy 117.0 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	Lena Lighting	100857	MIMO 2 LED BASIC 1510mm 4550lm IP66 840 (33W)	38.9 W	4550 lm	117.0 lm/W

Carpark · GROUND FLOOR · STORAGE (Light scene 1)

Calculation objects



Carpark · GROUND FLOOR · STORAGE (Light scene 1)

Calculation objects

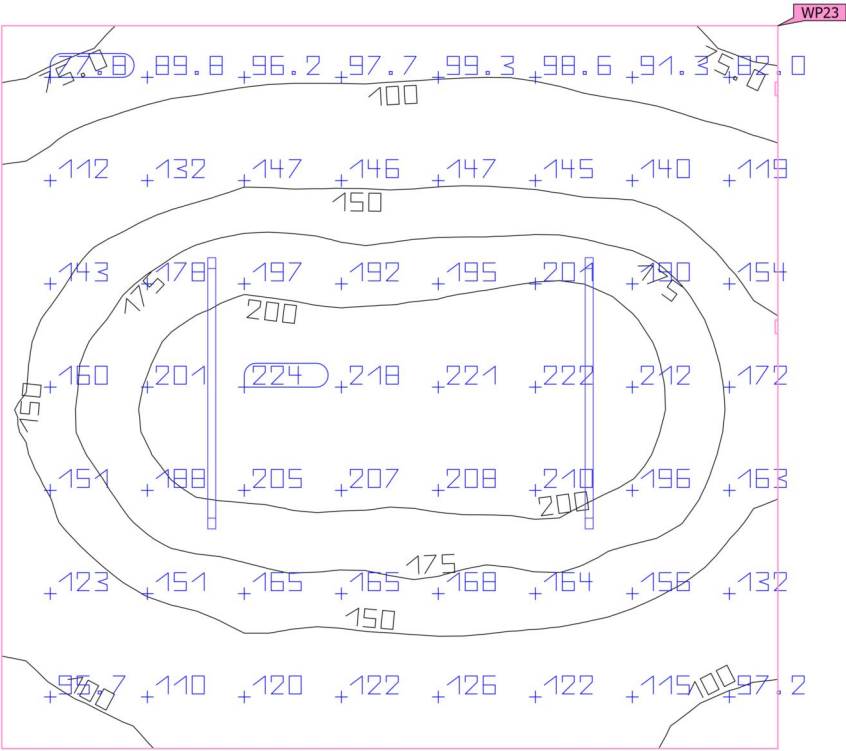
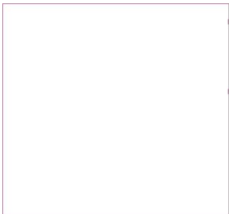
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (STORAGE) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.000 m	153 lx (≥ 80.0 lx) ✓	66.2 lx	225 lx	0.43 (≥ 0.40) ✓	0.29	WP23

Utilisation profile: General areas inside buildings - Store rooms, cold stores (12.1 Store and stockrooms)

Carpark · GROUND FLOOR · STORAGE (Light scene 1)

Working plane (STORAGE)



Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (STORAGE)	153 lx	66.2 lx	225 lx	0.43	0.29	WP23
Perpendicular illuminance (adaptive)	(≥ 80.0 lx)			(≥ 0.40)		
Height: 0.800 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: General areas inside buildings - Store rooms, cold stores (12.1 Store and stockrooms)